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Characteristics of an Acoustic Community: Puerto Rican Frogs of the Genus *Eleutherodactylus*

GEORGE E. DREWRY AND A. STANLEY RAND

The advertisement calling of 14 of the 15 species of *Eleutherodactylus* known from the main island of Puerto Rico was studied. When and where the species call and the characteristics of their advertisement calls are described. Calling assemblages of 4 to 6 species were found to be common, assemblages of more than 8 species were rare. The maximum numbers of species heard calling in one assemblage were at mid-elevations, at the border between forest and open habitat. Within an assemblage calls of different species differ in both dominant frequency and temporal patterning (duration and number of notes and durations of intervals). Potentially, differences in both frequency and temporal pattern in these frogs may contribute to species recognition and to the minimization of interference between the calls of different species in the same assemblage.

A vocalization evolves as the result of a variety of selection pressures. Some of these are generated by the physical nature of the communicating animals, others by the function of the vocalization and still others by the environments in which it is used.

Early work summarized by Blair (1958, 1964), Bogert (1960) and Littlejohn (1959), emphasized the importance of calling in the evolution and maintenance of reproductive isolation in frogs and this approach has received subsequent attention from many workers who have concerned themselves with both taxonomic and ecological assemblages (Elzen and Kreulen, 1974; Duellman, 1967; Duellman and Pyles, 1983; Hodl, 1977; Littlejohn, 1959, 1969; Schiøtz, 1967).

Subsequently, Elzen and Kreulen (1979), Littlejohn (1977), Littlejohn and Martin (1966, 1969), Loftus-Hills and Johnstone (1970) and Rivero (1978) have pointed out a related but separate kind of selection pressure on anuran communication systems, i.e., the need to avoid acoustic interference.

Selection for strategies that minimize interference acts along with selection for effective premating isolating mechanisms. The latter favors any perceptible differences between calls, especially those of closely related species. The former favors the reduction of interference regardless of phylogenetic relation or mating possibility and can operate across ordinal or class lines as well as against interference of nonbiological origin. Other factors such as acoustically

hunting predators and the transmission characteristics of the environment may also influence selection on frog calls.

In this paper we describe the spectral and temporal characteristics of the advertisement calls of 14 species of *Eleutherodactylus* on Puerto Rico, and consider how these properties function to 1) identify the species of the calling male and 2) reduce acoustical interference in mixed species choruses.

There are 15 species of *Eleutherodactylus* on the main island of Puerto Rico, all native. Various aspects of their morphology and acoustic biology have been described (Narins and Capranica, 1976, 1978, 1980; Narins and Hurley, 1982; Schmidt, 1928; Grant, 1931, 1932; Schwartz, 1969; Rivero, 1956, 1963, 1978; Rivero et al., 1963; Thomas, 1966). Rivero (1978) includes a phonograph record of their calls. Only one of these species, *E. jasper* (Drewry and Jones, 1976) is not included in this paper. Frogs in the genus *Eleutherodactylus* lack a free-swimming tadpole stage and, therefore, the frogs do not gather at aquatic breeding sites; instead, advertisement calling and probably pair formation take place within the normal home range of the male. Since the other species of frogs in Puerto Rico call from waterside breeding sites, we can, by studying *Eleutherodactylus*, study the entire amphibian contribution to most of the sonic environments in which *Eleutherodactylus* participates. The only other amphibian whose calls are regularly heard with those of *Eleutherodactylus* is *Leptodactylus albilabris*.

MATERIALS AND METHODS

Five data sets are considered here:

1. A series of tapes recorded in June 1969 using a Nagra II recorder and Sennheiser omnidirectional 104 condenser microphone; at localities listed in Table 1 for analysis of frequency and temporal patterns. For individual calls the microphone was placed as close as practical to a calling animal, usually within 0.5 m (sonagrams in Fig. 1 come from these recordings, as does the data in Table 2). A few minutes of the sound produced by the assemblage was also recorded at each locality with the microphone located so that the recording was not dominated by any one individual (Fig. 2).

2. A series of temporal patterns was selected as representative from tapes including more than 20 min of undisturbed calling by individual males (Fig. 3). These tapes were recorded in late 1969 and the first half of 1970 using a variety of recorders and microphones having dependable tape speed but less rigid frequency response specifications than the Nagra-Sennheiser combination.

3. A transect series of assemblages was taped 12 Oct. 1969, along Puerto Rico Highway 191, which transects the north face of the Luquillo Mountain range, using a Uher 4000L tape re-

corder with a Uher omnidirectional microphone (Fig. 4). The microphone was held approximately 2 m above the center of a traffic-free road. Each station was estimated to include calls from more than 20 frogs (mostly *E. coqui* at lower elevations), none closer than 10 m. These tapes were used to examine the relationship between elevation and call frequency (Fig. 5).

4. Diel activity cycles of *Eleutherodactylus coqui*, *E. antillensis*, *E. hedricki*, *E. brittoni* and *E. eneidae* recorded at the El Verde Field Station in 1969 and 1970 using a tuned amplifier, trigger circuit and recording ratemeter (Drewry, 1970).

5. Field notes regarding range, habitat and diel and seasonal patterns in calling activity of all species, collected by one of us (GED) during nearly continuous residence at El Verde Field Station (350–450 m altitude) from 1964 to 1972. Data sets 4 and 5 are not reported in detail here but were used to construct a summary of range and microhabitat distribution (Table 1) and the graph of seasonally averaged diel patterns (Fig. 6) included here as qualitative background information.

Frequency and temporal relationships in individual calls were measured from audiospec-

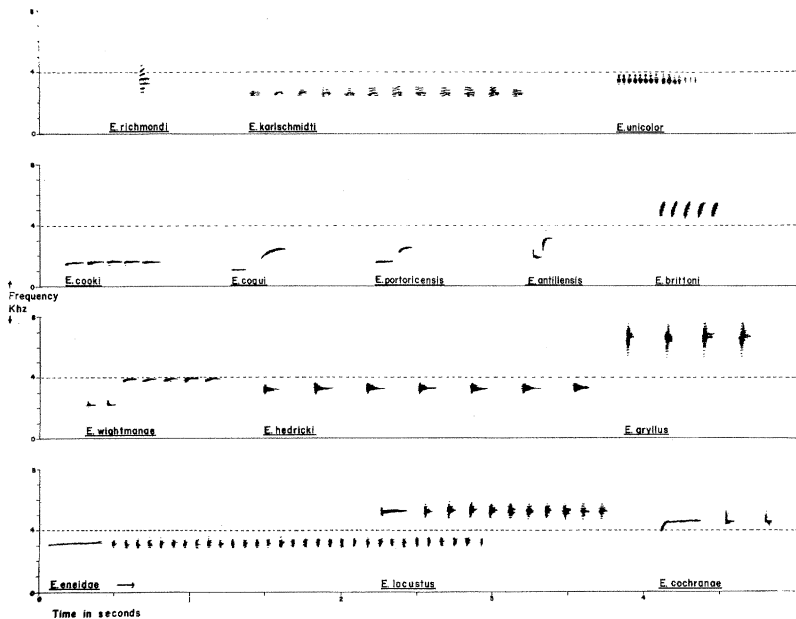


Fig. 1. Audiospectrograms of representative calls for 14 of the 15 Puerto Rican *Eleutherodactylus*.

TABLE 1. RECORDING LOCALITIES AND SPECIES HEARD, JUNE 1969.

Locality	Elevation	Time	Species of Eleutherodactylus ¹													
			R	U	K	CQ	P	A	B	CK	W	H	E	L	CR	G
1. Coffee grove, Mayaguez Hilton Hotel, Mayaguez	25 m	2300 hr				CQ		A	B							
2. E. Verde Field Sta., Hwy 186, km 19.3, 3 km S of El Verde	450	1900–0400	R		K	CQ	P	A	B		W	H	E			G
3. Campus, UPR Regional College at Cayey	425	1830–2000				CQ		A							CR	
4. Roadside, Hwy 15, 6.5 km SE Cayey	750	2020–2100				CQ	P				W			L		G
5. Roadside, Hwy 15, 10 km NW Guayama	300	2130–2145	R					A							CR	
6. Roadside, Hwy 3, 6.5 km SW Yabucoa (Panduras Mts.)	250	0130–0230				CQ				CK						
7. Roadside, Hwy 186, 13 km SSE El Verde	600	0430–0500	R		K	CQ	P		B		W	H	E	L		G
8. ElYunque Rock, near ElYunque Peak, Luquillo Mts.	1,025	2300–2400	R	U		CQ	P							L		G
9. Roadside meadow, Hwy 3, in Carolina	10	1900–2100				CQ		A	B						CR	

¹ R = richmondi, U = unicolor, K = karlschmidti, CQ = coqui, P = portoricensis, A = antillensis, B = brittoni, CK = cooki, W = wightmanae, H = hedricki, E = eneidiae, L = locustus, CR = cochranae, G = gryllus.

TABLE 2. CALL CHARACTERISTICS OF PUERTO RICAN *Eleutherodactylus*. No corrections were made for altitude or temperature.

Species	Sample N	Kinds of notes per call	Dominant Frequency KHz	No. of Notes per call	Note length msec	Note interval msec
1. <i>E. richmondi</i>	26	1	3.0–5.0	1 or 2	30–70	55–80
2. <i>E. unicolor</i>	10	1	3.5–3.8	8–10	8–25	30–40
3. <i>E. karlschmidti</i>	6	1	2.6–3.0	12–25	65–80	145–225
4. <i>E. coqui</i>	34	2	1.2–1.9 1.6–3.5	1 1	65–120 80–200	170–265
5. <i>E. portoricensis</i>	15	2	1.5–1.8 2.2–3.0	1 1	50–80 50–80	95–200
6. <i>E. antillensis</i>	31	2	1.8–2.2 2.5–3.6	1 1	25–65 50–90	40–65
7. <i>E. brittoni</i>	18	1	4.5–6.0	2–6	40–70	65–90
8. <i>E. cooki</i>	11	1	1.5–1.8	3–7	80–120	120–160
9. <i>E. wightmanae</i>	20	1 or 2	2.2–2.5 3.3–4.0	0–2 4–8	40–60 50–80	95–190 95–190
10. <i>E. hedricki</i>	18	1	2.9–3.4	7–19	80–120	310–410
11. <i>E. eneidae</i>	15	2	3.0–3.5 3.0–3.5	1 17–33	295–400 10–15	385–440 70–95
12. <i>E. locustus</i>	13	2	4.5–5.3 4.5–5.3	1 10–16	135–240 15–25	225–400 120–320
13. <i>E. cochranae</i>	34	1 or 2	3.7–4.8 3.7–4.8	1 0–3	230–305 15–20	335–465 225–265
14. <i>E. gryllus</i>	22	1	6.7–8.0	1–5	25–80	225–375

trograms made using a Kay Electric Model 6061A Sound Spectrograph. Call assemblages were studied in audiospectrograms and with a Bruel and Kjaer Model 2107 Frequency Analyzer set at 44dB per octave selectivity. The time constant of the latter instrument was extended to more than one minute for transect frequency analysis by adding 20,000 microfarads of capacitance across its meter circuit. Sounds to be analyzed were recorded on a ten-second tape loop and repeated into the analyzer at stepped frequency settings until the meter stabilized. Each point in Figs. 4, 5 thus represents the contribution at one frequency from all sounds occurring in the 10 second interval, integrating the calls of many separate animals. For temporal call pattern analysis of individual frogs, audio output of the 2107 analyzer was tuned to a dominant or distinctive frequency of the animal under study, rectified and smoothed in a simple diode-resistor-capacitor network, and used to drive a pen recorder. Such pen recordings of representative frogs were traced to produce Fig. 3.

For each species we have described 1) typical calling site; 2) typical peak calling times in the diel and seasonal cycles; 3) typical short-term temporal patterns in calling, e.g., note duration and repetition rate, number of notes per call, combinations of long and short notes, call rates and characteristic changes in rates of calling; 4) modulation patterns of frequency and amplitude; 5) dominant frequency; and 6) call bandwidth related to frequency and/or amplitude modulation. A note was defined as the sound resulting from a single contraction of the thoracic musculature. Calls were defined as the unit of repetition where repeating sequences of dissimilar note frequencies, durations or spacing were identified. The frog calls that we could hear while standing in one place were considered to constitute an assemblage.

RESULTS

Spatial patterns.—Species differed in the places from which they called. Three levels of spatial differences were noted: geographic, microhab-

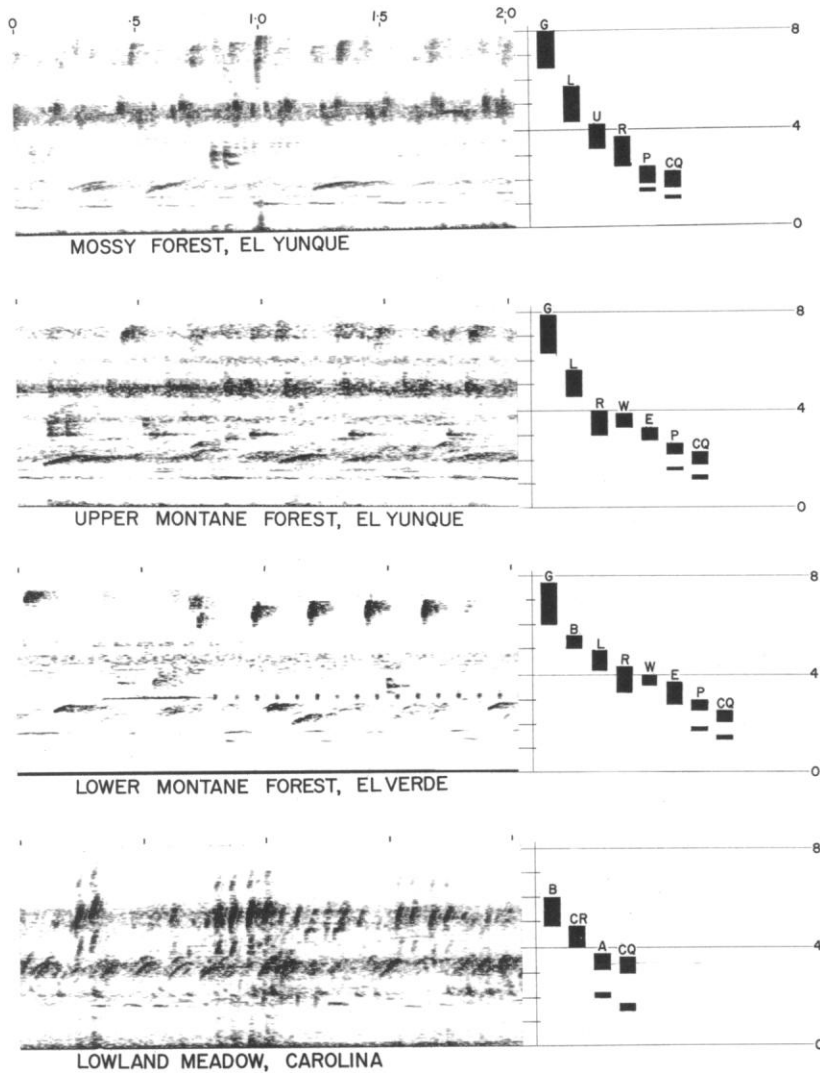


Fig. 2. Calling assemblage audiospectrograms of Puerto Rican *Eleutherodactylus*. Relative spectral distributions diagramed at right. Species abbreviations as in Table 1. Unlabeled bands were produced by insects.

itat, and call site (Table 3). Species that differed in geographic distribution in Puerto Rico most commonly also differed in the altitudes at which they occurred. Interspecific interactions may hinder two allopatric species with similar calls from becoming sympatric or, where they are sympatric, act to keep them in different microhabitats. An interesting pair of species pairs in this regard are *E. locustus* and *E. brittoni*, which have overlapping dominant frequencies and areas of both sympatry and allopatry. In allo-

patry they occupy similar microhabitats, but, in the narrow bands along road edges where they are sympatric, *E. locustus* is almost entirely restricted to fern thickets and *E. brittoni* to grassy areas.

Call site differences were noticeable among many species (Table 3). The clearest separation involved the species restricted to streams, *E. karlschmidti*, but even this species was found in close proximity to others in tiny brooks inside the forest. There was a tendency observed in

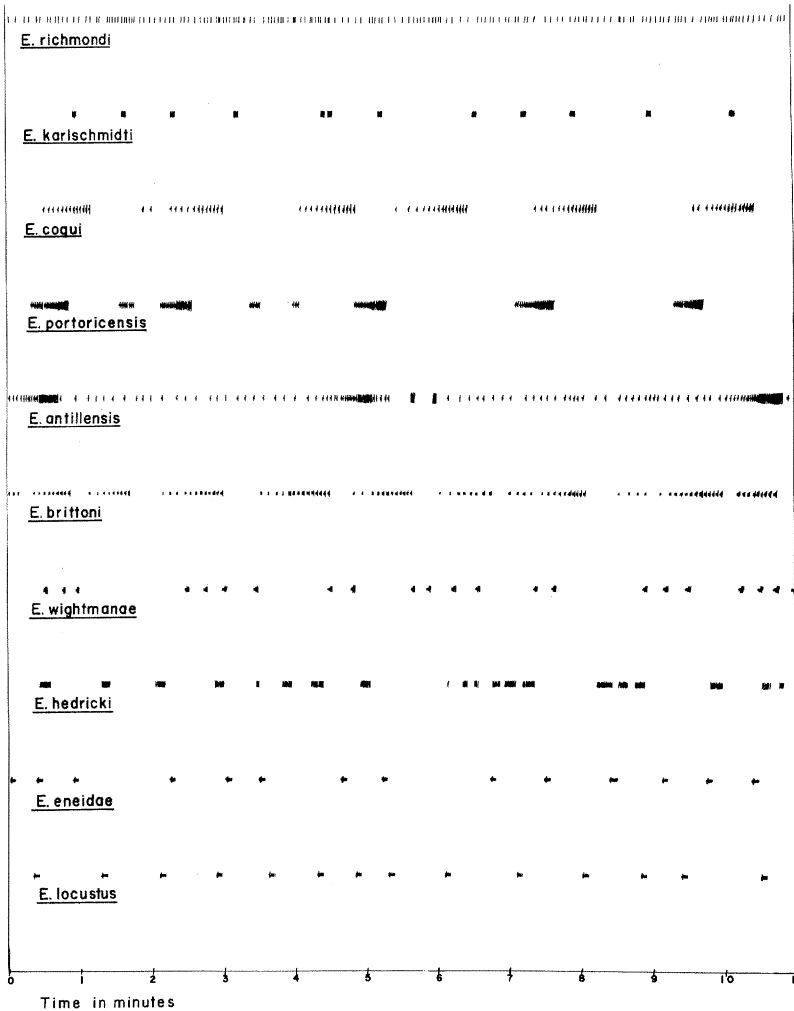


Fig. 3. Typical patterns of calling during a seventeen minute time interval for Puerto Rican *Eleutherodactylus*. Reproduced from chart recordings.

all species, except *E. karlschmidti*, *E. unicolor* and *E. hedricki*, to move upward a short distance during the course of the evening (Schwartz, 1969). Thus the terrestrial species, *E. wightmanae* and *E. richmondi*, which began calling beneath the litter, were observed to climb to horizontal stems and leaves 15 to 100 cm above ground, and the other species which use arboreal sites were seen to move upward while retaining the same relative spatial distributions.

During our recording, we seldom heard more than eight species at any one site. For a given habitat, four to six species were more typical,

but along ecological boundaries species from two habitats were commonly audible and these sites yielded the most complex assemblages. At middle elevations *E. locustus* and *E. gryllus* were confined to ecotonal boundaries along forest edges and did not extend far into either forest or meadow.

We identified only a forest assemblage in high elevation mossy forest and in the upper montane forest just below it. At the elevation of middle and lower montane forest (100–600 m), two assemblages were heard: one in meadows which included predominantly *E. antillensis* and

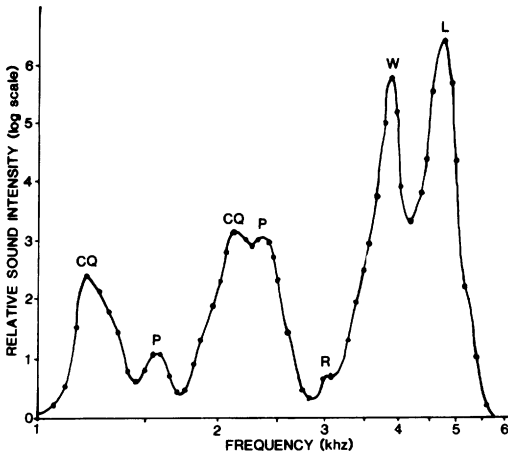


Fig. 4. Frequency spectrum distribution in an upper montane forest assemblage (B & K analyzer). Each point represents intensity averaged over one minute. Scale factor chosen arbitrarily. Species abbreviations as in Table 1.

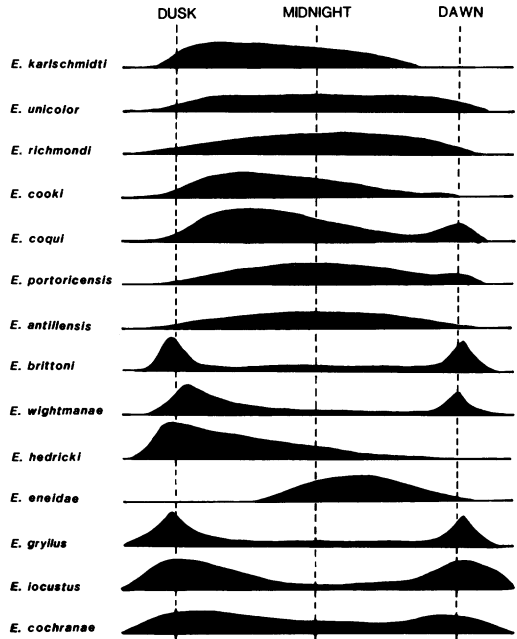


Fig. 6. Diel patterns of vocal activity for each species. Vocal activity (the number of calls heard per unit time) is scaled for each species so that maxima are about equal. Patterns shown can be modified by several factors such as moisture, temperature and season (Drewry, 1970).

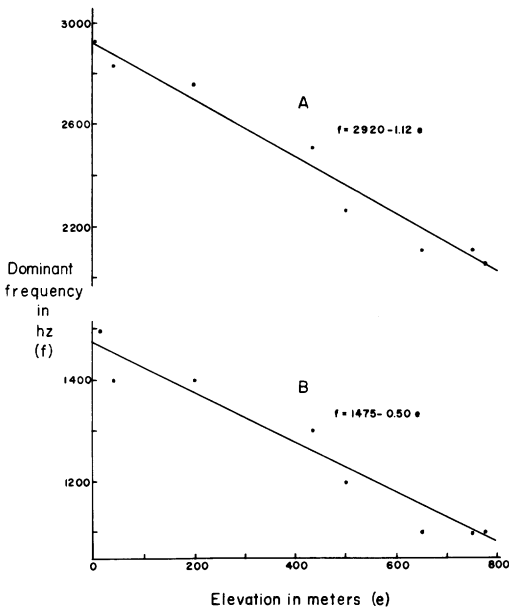


Fig. 5. Relationships between elevation and dominant frequencies in *E. coqui*. A = high-pitched second notes. B = low-pitched first notes. Points represent frequency peaks of choruses taken from plots such as that in Fig. 4. Scale factor of A and B are different, illustrating geometric frequency relationship.

E. brittoni, and another in the forest where these two species were seldom audible but *E. hedricki* and *E. portoricensis*, strictly forest species, could be heard. The assemblage chosen to represent lower montane forest in Fig. 2 included *E. brittoni* and *E. portoricensis*, but lacked *E. antillensis* and *E. hedricki*. Therefore, it is clearly a forest-edge assemblage, and illustrates what is probably about the maximum acoustic complexity to be heard anywhere on the island.

Temporal patterns.—At the El Verde Field Station we observed both diel and seasonal differences in times of *Eleutherodactylus* calling. During the cooler evenings in the winter, when temperatures often dropped to 16–17 C, the only species calling actively by 2100 hr were *E. richmondi*, *E. karlschmidti*, and, at higher elevations, *E. unicolor*. Fig. 6 shows relative diel activity cycles in *Eleutherodactylus* calling when temperatures are above about 19 C. It is based on a combination of our field notes and ratemeter

TABLE 3. RANGE, MICROHABITAT AND CALL SITES IN PUERTO RICAN *Eleutherodactylus* (FROM NOTES BY G. DREWRY AND J. A. RIVERO, PERS. COMM.)

Species	Distribution	Microhabitat	Calling site
<i>richmondi</i>	Montane forest above 150 m	Forest floor, stone crevices, under logs etc.	Crevices, sometimes climbs plants 2–600 cm
<i>unicolor</i>	Eastern montane forest above 600 m	Subterranean burrows in root plexus, under stones etc.	2–7 cm below litter surface
<i>karlschmidti</i>	Montane streams above 150 m	Stream boulders, usually near waterfalls	Any height on or between boulders
<i>coqui</i>	Throughout island, no altitude limits	Arboreal on trees, shrubs and dwellings	1–15 m above ground
<i>portoricensis</i>	Montane forest above 180 m	Forest litter by day, climbs at night	.5–5 m frequently on young palms
<i>antillensis</i>	Throughout island below 1,000 m	Shrubs, forbs, grass, usually in sunny areas	0–1 m seldom on grass stems
<i>brittoni</i>	Throughout island below 700 m	Forbs, tall grass in meadows, usually sunny areas	0–1.5 m often on grass stems
<i>cooki</i>	Southeastern mts. only. 120–300 m	Crevices in granitic boulders	Any height on boulders, often subterranean
<i>wightmanae</i>	Montane forest above 150 m	Forest litter layer, female seldom climbs	Litter surface, sometimes climbs plants to 30 cm
<i>hedricki</i>	Montane forest above 270 m	Arboreal, almost restricted to treeholes	Within or near male's treehole, 1–15 m
<i>eneidae</i>	Montane forest above 270 m	Banks, boulders, logs or trunks, often in or near moss	Mossy surfaces, crevices, sometimes climbs plants 1–1.5 m
<i>locustus</i>	Eastern montane forest above 370 m	Forest edge, mossy forest meadows above 700 m	0–1.5 m, frequently on ferns
<i>cochranae</i>	Costal plain up to 370 m	Marshes, lawns, meadows, bromeliads in dry forest	0–2 m on grass or weeds, to 8 m in dry forest
<i>gryllus</i>	Montane forest above 370 m	Forest edge, bromeliads in mossy forest, banana groves	Usually 1–4 m above ground

recordings for several species (see Materials and Methods). Exact timing can be modified by season, rain, wind and other weather variables (Drewry, 1970).

Figs. 1 and 3, in conjunction with Table 2, illustrate that, on the basis of length and number of notes and intervals between notes and between calls, each species has a call with a unique temporal pattern. Each species is represented in Fig. 1 by a representative note or note sequence considered to be a single call. Notes of *E. hedricki* could be alternatively in-

terpreted as separate calls, but note spacing is relatively invariant, even though number of notes per call is rather variable. Fig. 3, where calls are represented as single marks, shows that all species vary call spacing to a greater or lesser degree. The number of notes in a typical call is variable in all species producing more than two notes per call (Table 2).

Call sequences shown in Fig. 3 are representative. Patterns differ somewhat between individuals, with time of night and with the social situation. These patterns are least clear in early

evening or just after showers, when individuals are calling most nearly continuously. A distinctive pattern, which we term ramping, is well developed in the three species with two-frequency calls, *E. coqui*, *E. portoricensis* and *E. antillensis*, and in *E. brittoni* and *E. unicolor*. In this ramping pattern calls follow one another more and more rapidly, gradually accelerating to a peak rate and then abruptly slowing or stopping. In *E. brittoni* the number of notes per call increases gradually from two to five or six in the course of each ramp. Ramp peaks in *E. antillensis* are frequently, but not always, followed after about 30 sec. by a different call (indicated in Fig. 3 by a wide mark), higher in pitch than the usual calls and consisting of a sequence of clicks. A similar occasional call is often given by *E. coqui* males in the daytime or very early evening. A similar but lower-pitched sequence is the usual call of *E. cooki*, which also exhibits a weak pattern of ramping. The pattern shown in Fig. 3 for *E. richmondi* differs from typical ramping in that the deceleration in repetition rate is seldom more abrupt than is the acceleration. The functional significance of these ramping patterns is unknown.

Frequency patterns.—Bandwidth: Compared with most hylids, rhacophorids and leptodactylids for whose calls audiospectrograms have been published (Duellman, 1970; Schiøtz, 1967; Straughan and Heyer, 1976) most Puerto Rican *Eleutherodactylus* have calls with a narrow bandwidth, though none uses a single frequency and several use a wider band of frequencies.

A greater bandwidth can be the result of sidebands produced by amplitude modulation. The notes of two Puerto Rican species show amplitude modulation which is consistently 100% (Littlejohn, 1971); *E. richmondi*, with a pulse rate near 200 pps., and *E. karlschmidti*, with a rate of about 100 pps. Modulation sidebands are visible in audiospectrograms of the calls of these two species (Fig. 1). The subjective impression they produce is of harshness of the notes, and one consequence is the widening of spectral bandwidth. Amplitude modulation near 100% is exhibited by *E. unicolor* and *E. gryllus*, whose calls are therefore harsh, but pulse rate varied and could not be measured usefully with our sample. The short notes of several other species (e.g., *E. eneidae*, *E. locustus* and *E. cochraniae*) show some amplitude modulation with consequent increase in bandwidth and subjective harshness.

Five Puerto Rican species, *E. coqui*, *E. porto-*

ricensis, *E. antillensis*, *E. brittoni* and *E. wightmanae* achieve a greater bandwidth by a significant frequency modulation in their calls. Unlike the calls whose broad bandwidth is produced by amplitude modulation, these do not sound harsh. The first four of these species exhibit frequency sweeps within the course of single notes and all but *E. brittoni* jump within a single call from one dominant frequency to another, thus contributing two frequency bands to the audio spectrum of the assemblage. The first three species also produce occasional calls of slightly higher frequency and several notes per call. All of these species are very common, and at least two were found to be predominant elements in almost every assemblage selected for study. Although each of the first three species contributes two distinct spectral bands or frequency channels, the upper band of each species overlaps widely with the others, while the first notes of each are narrowly tuned, not frequency modulated, and fall within a restricted band that is species-specific (Fig. 2).

Frequency separation: In one place and at one time, the species that are calling in the assemblage use different frequency bands with little overlap between them (Figs. 2, 4). There are exceptions (*E. richmondi* overlaps several other species, and, as noted above, parts of the calls of *E. antillensis*, *E. coqui* and *E. portoricensis* overlap one another), but frequency separation between the calls of the species at any one locality is the general rule.

A significant fraction of the variability in dominant frequency shown in Table 2 was correlated with the different altitudes at which the samples were recorded. Typically a species at one elevation uses a narrow frequency band. In some species there is a negative correlation between altitude and dominant frequency, and this in turn seems to be correlated with larger size of the animals at higher elevations. Fig. 5 shows this relationship in *E. coqui*. The points in Fig. 5 represent averaged frequency peaks from analyses of tapes such as those shown in Fig. 4.

In all of the relatively narrow-tuned species there was considerable individual variation in dominant frequency so that the characteristic bandwidth for each species within an assemblage was about 0.5 KHz (Fig. 2).

DISCUSSION

We have described the advertisement calls of 14 of the 15 *Eleutherodactylus* species on the main

island of Puerto Rico in terms of the differences between them in where and when they are given and in their physical properties. We will now consider how these differences may function to insure species recognition and reduce acoustic interference between species.

Temporal and spatial differences in call site have been commonly noted between sympatric frogs and suggested to function as isolating mechanisms (Duellman, 1967). They also reduce interference, functioning in this regard even when the separation is only partial. However, among Puerto Rican *Eleutherodactylus* the great overlap in spatial distribution and in time of calling (Table 1) means that other mechanisms of species recognition and of reduction of interference between calls must be important.

The various calling rates and combinations of long and short notes result in each species having a unique temporal pattern. The complexity of these temporal patterns suggests that temporal uniqueness of a call is an important requirement for an effective call.

Temporal patterning is important in species recognition in frogs. The experiments of Loftus-Hills and Littlejohn (1971) and Gerhardt (1978) demonstrate that females of *Hyla* species can discriminate between synthetic calls which differ only in pulse rate, choosing the rate typical of their species.

Temporal patterns themselves seem to contribute little to the avoidance of interference, and, in fact, any addition to the length of a call, or proportion of time calling necessary to achieve uniqueness increases the probability of interference. The primary function of temporal patterns appears to be in species identification. If frogs used signal averaging techniques they could recover temporal patterns in the presence of noise. Hodl (1977) pointed out the difficulties of recognizing temporal patterns in large choruses of the Amazonian frogs and stressed the importance of frequency differences in discriminating species at any distance.

The observed differences among the Puerto Rican *Eleutherodactylus* in dominant frequencies are not fortuitous consequences of differences between species in body size. The largest species in the fauna is *E. karlschmidti*, yet its call is higher in pitch than those of at least four smaller species. The high dominant frequency in this species is probably related to the unusual morphology of the vocal pouch that, in this species, is divided into two relatively small sacs centered on lateral

vocal sac openings (Rivero et al., 1963; Schwartz, 1969). This high-pitched call may be an adaptation to the noisy mountain streams along which this species lives. The general trend of higher pitched calls in smaller frogs is also reversed by *E. richmondi*, whose call is higher than that of the smaller *E. eneidae*, and by *E. unicolor*, whose call is much lower than several other species of its general size.

We have shown that generally, within an assemblage, different species are using different frequency bands in their advertisement calls. Frequency overlap between the calls of heterospecifics should shorten the distance over which conspecific calls can be received and recognized, regardless of the uniqueness of their temporal patterns. This should generate a selective advantage to reducing interference by calling at a unique frequency. Differences in the dominant frequencies of frogs calling in the same assemblage should function to reduce interference between them.

What is the role of frequency in species recognition? We agree with Loftus-Hill and Johnstone (1970:1136) that "the extent to which mating call frequency differences carry information relevant to species identification is not clear." Bullfrogs (Capranica, 1965) and green treefrogs (Gerhardt, 1974, 1976) both respond best to synthetic calls that have frequencies represented at the same relative amplitudes that they are in natural calls.

Frogs evidently have tuned ears from which the majority of neurons carry responses to a narrow frequency range of biological interest (Capranica, 1977). With probably somewhat less ability to identify and thereby reject "wrong" frequencies than vertebrates possessing cochleae, frogs would be expected to rely heavily on temporal coding for specificity. Gerhardt (1978) showed that green treefrogs respond positively to noise coded in the temporal pattern of natural calls, indicating that presence of many unnatural frequency components did not destroy the signal. However, Gerhardt (1974) also showed that synthetic calls with components at certain frequencies were less effective than synthetic calls containing only the normally emphasized frequencies. Further, he reports (Gerhardt, in press) that female *H. cinerea* and *H. gratiiosa* can discriminate between two synthetic calls differing only in frequency, each selecting the call with the dominant frequency characteristic of their species. However, at least in *H. cinerea*, the discrimination is fragile and disap-

pears if the call at the heterospecific frequencies is more than 12 db louder. Until the ability to utilize frequency to identify calls has been investigated in the Puerto Rican *Eleutherodactylus* we can only speculate that the frequency differences between species are relatively more important in interference avoidance than in species identification. Most probably they are important in both.

The function of the differences in bandwidth, whether achieved by frequency or amplitude modulation, is not at all clear. If a call has its energy concentrated in a single, narrow frequency band, and the receiving ear is tuned to the same frequency, the result should be the system of maximum efficiency. Why then do certain Puerto Rican species, and a great many species of frogs in other areas, have wide-band calls? One possible explanation for the band spreading associated with amplitude modulation is that it is an unavoidable consequence of temporal coding of the call by breaking it into pulses. Another possibility is that if bandwidth of the receiving ear matches that of the transmitter, a harsh, wide-band call, like that of *E. karlschmidti*, may be less susceptible to jamming by interfering sounds than are narrowly-tuned calls. This species generally occupies a high noise environment along torrential streams.

Band spreading by frequency modulation could also sidestep interfering frequencies from other frogs, a communication strategy potentially available to a few species in any complex fauna. The frequency-modulating species in Puerto Rico, however, achieve overlap rather than distinctiveness in the frequency-modulated portions of their calls and all but *E. brittoni* exhibit a separate, species-specific band in which frequency is not modulated. The findings of Narins and Capranica (1976) that *E. coqui* males are more responsive to the unmodulated, species-specific note of the two-note call and the females more responsive to the second frequency-modulated note which overlaps more with the calls of other species are surprising. We would expect reproductively motivated females to be more responsive to the species-specific notes than to the notes that overlap more the notes of other species.

Differences among the calls of sympatric assemblages of species of frogs have been described for several places in the tropics: Puerto Viejo, in the Atlantic lowlands of Costa Rica; Santa Cecilia, in Amazonian Ecuador; Belem, on the lower Amazon (Duellman and Pyles,

1983); Manaus, in the middle Amazon (Hodl, 1977); and the Serengeti, in East Africa (Elzen and Krueken, 1979). These published accounts report that the species calling together all differ from one another in call, while species that are geographically separate may, but do not necessarily, have very similar calls. Most species in an assemblage have different dominant frequencies and also different temporal patterning. Those species similar in one of these domains usually differ greatly in the other and often also in calling site or season.

Puerto Rican *Eleutherodactylus* seem to have more species with narrow-band calls and within assemblages better frequency separation between species, even though each species also has a unique temporal pattern. This cleaner, or more redundant separation among species of *Eleutherodactylus* in Puerto Rico may be related to the very high population densities of many of these species (Turner and Gist, 1970) and perhaps to their breeding biology. Because of their direct development, these frogs do not gather at breeding ponds to call but call from their normal home ranges.

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Life History Variation among Experimental Populations of the Mosquitofish, *Gambusia affinis*

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The life history variation of the poeciliid, *Gambusia affinis*, was examined in a series of experimental ponds. Fish from a single field location near Perth, Western Australia, were maintained for a summer breeding season at five different food levels. The ponds were used to establish if the life history response of the fish to an environmental continuum was based upon consistent relationships among the traits and close covariation of the life history as a whole. Principal component analysis was used to define the covariation of the traits.

The fish initially responded to the increasing food by increasing somatic and reproductive growth. However, somatic growth reached a maximum at intermediate food levels although fecundity and brood mass still increased at all food levels. Embryo mass varied little. The principal component analysis identified three independent sources of variation. The first component, which was interpreted as defining the covariation of the traits in response to the food continuum, was only closely associated with fecundity and brood mass.

The results indicate that the covariation of the traits is not defined by consistent relationships and that the response of the fish to the food is not adequately described in terms of a simple life history "tactic." This is partly because there are differences between the traits in their ability to vary. Explanations of the life history variation must take into account that, even given relatively simple environmental variation, there may be more than one dimension to the life history response and that these cannot be defined by exclusive sets of traits.

LIFE history traits such as fecundity and age at maturity have in common a direct influence on the amount and timing of reproduction. These traits are generally regarded as a coadapted set (Stearns, 1976), in which it is the combination of traits that is important in defining a particular life history response to an environment. Close relationships among the traits are implicit in this idea of a life history, and the more a life history responds to the environment as a coherent set of traits the more closely will the traits covary.

In a recent paper (Trendall, 1982) I exam-

ined the covariation of life history traits in a number of field populations of the mosquitofish, *Gambusia affinis*, and came to the conclusion that the covariation of the traits did not support the assumption that the life history responded as a single, coadapted unit or that fixed relationships among the traits were an integral part of life history organization.

It is often the case that observed life history variation in field populations does not support theoretical explanations (Fleming, 1979; Horn, 1978; Schaffer, 1979; Stearns, 1977; Wilbur et al., 1974), and in a comprehensive review of life